Table 2: Coverage (%) − Shrinkage

Scenario I: all means are 0 for T0,T1,...,T5

	None adjustment						Many−to−one adjustment					
Missing rate	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**Joint**	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**Joint**
0%	94.99	95.09	95.06	95.09	95.46	82.50	98.84	98.63	98.70	98.46	98.59	94.35
Imputation using linear regression												
10%	95.14	94.77	94.68	94.79	94.41	81.08	98.82	98.70	98.47	98.65	98.48	94.45
20%	95.29	95.14	95.17	95.01	95.10	82.82	98.91	98.86	98.84	98.72	98.54	95.07
30%	95.62	95.77	95.74	95.58	95.66	84.30	99.12	99.05	99.10	99.10	98.79	95.99
40%	96.19	95.87	95.97	96.06	95.93	85.37	99.16	99.08	99.18	99.14	99.17	96.62
Imputation using predictive mean matching method												
10%	93.95	93.66	93.79	93.78	93.47	77.94	98.44	98.26	98.18	98.34	98.01	93.01
20%	92.98	92.80	93.02	92.38	92.97	75.70	98.01	97.88	97.95	97.74	97.69	91.69
30%	91.27	91.70	91.52	91.54	91.40	71.85	97.43	97.49	97.19	97.23	97.29	89.83
40%	89.91	89.78	89.95	89.82	89.69	67.62	96.73	96.46	96.46	96.23	96.20	86.95
Imputation using propensity−score method												
10%	93.95	93.53	93.65	93.83	93.29	77.75	98.34	98.21	98.15	98.23	97.88	92.71
20%	92.77	92.25	92.63	92.23	92.67	74.75	97.81	97.90	97.83	97.70	97.46	91.28
30%	90.85	91.25	91.34	91.39	90.99	70.94	97.10	97.37	96.89	96.95	96.91	88.97
40%	89.60	89.04	89.06	89.50	89.32	66.10	96.31	96.01	95.92	96.01	95.76	85.59

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

Table 2: Coverage (%) − Shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5

	None adjustment						Many−to−one adjustment					
Missing rate	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**Joint**	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**Joint**
0%	94.76	95.30	95.31	95.23	95.12	83.01	98.63	98.75	98.65	98.65	98.51	94.74
Imputation using linear regression												
10%	94.64	94.73	94.51	94.75	94.62	81.26	98.87	98.78	98.80	98.55	98.50	94.79
20%	95.19	94.95	95.10	94.82	95.36	82.72	98.88	98.84	98.79	98.65	98.89	95.30
30%	95.18	95.57	95.47	95.89	95.48	83.32	98.99	99.24	99.03	99.04	98.95	96.07
40%	95.79	95.87	95.69	95.88	95.73	84.70	99.05	99.00	98.84	99.27	99.00	95.98
Imputation using predictive mean matching method												
10%	93.58	93.72	93.44	93.72	93.65	78.11	98.53	98.33	98.34	98.07	98.11	93.17
20%	92.94	92.87	92.71	92.70	93.28	76.06	98.06	97.97	97.84	97.72	97.96	92.22
30%	91.75	91.42	91.49	92.06	91.66	71.77	97.30	97.55	97.32	97.44	97.36	90.07
40%	89.40	89.71	89.44	89.43	89.47	66.75	96.14	96.34	96.01	96.12	96.18	86.27
Imputation using propensity−score method												
10%	93.71	93.65	93.34	93.54	93.54	78.00	98.52	98.25	98.28	98.04	97.92	92.93
20%	92.63	92.53	92.78	92.53	93.16	75.40	98.02	97.76	97.85	97.62	97.98	91.82
30%	90.72	91.24	91.28	91.31	91.03	70.37	96.99	97.36	96.99	96.88	97.12	88.93
40%	89.14	89.00	88.68	88.88	88.92	64.93	96.10	95.87	95.74	95.61	95.43	84.68

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

Table 3: Average interval length (expansion %) − Shrinkage

Scenario I: all means are 0 for T0,T1,...,T5

	None adjustment					Many−to−one adjustment				
Missing rate	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**
0%	4.028	4.075	4.108	4.147	4.186	5.083	5.094	5.100	5.112	5.120
Imputation using linear regression										
10%	4.176 (3.67)	4.213 (3.39)	4.256 (3.60)	4.288 (3.42)	4.338 (3.63)	5.361 (5.47)	5.378 (5.58)	5.398 (5.84)	5.411 (5.87)	5.437 (6.19)
20%	4.539 (12.68)	4.573 (12.24)	4.617 (12.37)	4.662 (12.44)	4.711 (12.54)	5.824 (14.58)	5.842 (14.69)	5.873 (15.14)	5.901 (15.45)	5.939 (16.00)
30%	5.045 (25.23)	5.121 (25.68)	5.147 (25.29)	5.204 (25.50)	5.253 (25.48)	6.465 (27.18)	6.534 (28.27)	6.547 (28.36)	6.597 (29.07)	6.639 (29.68)
40%	5.825 (44.60)	5.855 (43.69)	5.909 (43.83)	5.983 (44.29)	6.056 (44.67)	7.447 (46.51)	7.470 (46.64)	7.524 (47.51)	7.599 (48.66)	7.675 (49.91)
Imputation using predictive mean matching method										
10%	3.997 (−0.78)	4.034 (−1.01)	4.074 (−0.83)	4.107 (−0.95)	4.154 (−0.77)	5.135 (1.01)	5.148 (1.07)	5.164 (1.25)	5.177 (1.28)	5.196 (1.49)
20%	4.110 (2.02)	4.143 (1.68)	4.183 (1.82)	4.224 (1.87)	4.266 (1.91)	5.280 (3.87)	5.294 (3.94)	5.317 (4.25)	5.336 (4.40)	5.364 (4.76)
30%	4.235 (5.12)	4.286 (5.17)	4.321 (5.18)	4.359 (5.11)	4.403 (5.18)	5.438 (6.99)	5.472 (7.42)	5.495 (7.74)	5.518 (7.96)	5.549 (8.38)
40%	4.376 (8.64)	4.413 (8.30)	4.450 (8.31)	4.500 (8.52)	4.549 (8.67)	5.614 (10.44)	5.641 (10.74)	5.672 (11.20)	5.712 (11.74)	5.752 (12.35)
Imputation using propensity−score method										
10%	4.007 (−0.52)	4.042 (−0.81)	4.084 (−0.60)	4.117 (−0.71)	4.164 (−0.53)	5.148 (1.28)	5.160 (1.30)	5.178 (1.51)	5.191 (1.56)	5.210 (1.77)
20%	4.137 (2.68)	4.169 (2.32)	4.211 (2.50)	4.250 (2.50)	4.294 (2.56)	5.314 (4.54)	5.328 (4.59)	5.353 (4.96)	5.372 (5.10)	5.402 (5.51)
30%	4.264 (5.84)	4.316 (5.93)	4.354 (5.97)	4.393 (5.93)	4.434 (5.91)	5.474 (7.68)	5.510 (8.17)	5.536 (8.53)	5.563 (8.83)	5.590 (9.18)
40%	4.365 (8.37)	4.401 (8.01)	4.447 (8.24)	4.496 (8.43)	4.548 (8.64)	5.597 (10.12)	5.625 (10.42)	5.666 (11.09)	5.709 (11.68)	5.752 (12.36)

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

Table 3: Average interval length (expansion %) − Shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5

	None adjustment					Many−to−one adjustment				
Missing rate	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**
0%	4.036	4.077	4.119	4.152	4.191	5.083	5.093	5.103	5.109	5.119
Imputation using linear regression										
10%	4.160 (3.07)	4.203 (3.09)	4.245 (3.08)	4.292 (3.38)	4.327 (3.25)	5.348 (5.21)	5.369 (5.43)	5.388 (5.59)	5.411 (5.91)	5.428 (6.03)
20%	4.545 (12.61)	4.600 (12.83)	4.636 (12.57)	4.692 (13.00)	4.732 (12.91)	5.831 (14.71)	5.869 (15.24)	5.892 (15.47)	5.929 (16.04)	5.957 (16.37)
30%	5.035 (24.75)	5.084 (24.69)	5.129 (24.54)	5.191 (25.03)	5.227 (24.72)	6.455 (26.98)	6.495 (27.52)	6.530 (27.96)	6.585 (28.88)	6.615 (29.22)
40%	5.821 (44.21)	5.885 (44.35)	5.921 (43.76)	5.978 (43.98)	6.040 (44.12)	7.439 (46.34)	7.505 (47.37)	7.531 (47.59)	7.591 (48.56)	7.652 (49.48)
Imputation using predictive mean matching method										
10%	3.986 (−1.24)	4.028 (−1.20)	4.067 (−1.25)	4.113 (−0.94)	4.145 (−1.10)	5.126 (0.83)	5.145 (1.02)	5.158 (1.07)	5.177 (1.33)	5.191 (1.40)
20%	4.119 (2.06)	4.165 (2.15)	4.198 (1.94)	4.249 (2.33)	4.284 (2.22)	5.290 (4.06)	5.313 (4.33)	5.332 (4.48)	5.358 (4.87)	5.378 (5.06)
30%	4.234 (4.90)	4.277 (4.91)	4.311 (4.67)	4.367 (5.17)	4.396 (4.89)	5.439 (7.00)	5.468 (7.37)	5.487 (7.52)	5.527 (8.18)	5.547 (8.36)
40%	4.354 (7.88)	4.404 (8.02)	4.446 (7.95)	4.485 (8.02)	4.527 (8.03)	5.590 (9.97)	5.631 (10.56)	5.661 (10.93)	5.690 (11.37)	5.724 (11.82)
Imputation using propensity−score method										
10%	3.997 (−0.97)	4.039 (−0.94)	4.077 (−1.01)	4.124 (−0.68)	4.156 (−0.82)	5.139 (1.10)	5.158 (1.28)	5.171 (1.34)	5.191 (1.60)	5.206 (1.70)
20%	4.145 (2.71)	4.189 (2.73)	4.226 (2.60)	4.278 (3.03)	4.310 (2.85)	5.322 (4.69)	5.344 (4.92)	5.365 (5.14)	5.396 (5.60)	5.413 (5.75)
30%	4.263 (5.61)	4.310 (5.70)	4.346 (5.51)	4.401 (6.00)	4.428 (5.66)	5.474 (7.68)	5.507 (8.13)	5.530 (8.36)	5.570 (9.02)	5.589 (9.19)
40%	4.357 (7.96)	4.404 (8.02)	4.465 (8.41)	4.492 (8.18)	4.541 (8.35)	5.591 (9.99)	5.629 (10.54)	5.683 (11.37)	5.700 (11.56)	5.740 (12.12)

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

Table 2: Coverage (%) − Non−shrinkage

Scenario I: all means are 0 for T0,T1,...,T5

	None adjustment						Many−to−one adjustment					
Missing rate	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**Joint**	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**Joint**
0%	94.99	95.09	95.06	95.09	95.46	82.50	98.50	98.48	98.52	98.43	98.61	93.80
Imputation using linear regression												
10%	95.14	94.77	94.68	94.79	94.41	81.08	98.63	98.42	98.38	98.64	98.46	93.85
20%	95.29	95.14	95.17	95.01	95.10	82.82	98.70	98.70	98.73	98.60	98.48	94.63
30%	95.62	95.77	95.74	95.58	95.66	84.30	98.88	98.87	98.88	98.96	98.75	95.33
40%	96.19	95.87	95.97	96.06	95.93	85.37	99.08	98.97	99.01	99.06	99.00	96.16
Imputation using predictive mean matching method												
10%	93.95	93.66	93.79	93.78	93.47	77.94	98.22	98.09	98.08	98.24	98.10	92.48
20%	92.98	92.80	93.02	92.38	92.97	75.70	97.67	97.73	97.74	97.55	97.52	90.74
30%	91.27	91.70	91.52	91.54	91.40	71.85	96.95	97.18	97.00	97.07	97.11	88.74
40%	89.91	89.78	89.95	89.82	89.69	67.62	96.15	95.98	96.13	95.85	95.88	85.33
Imputation using propensity−score method												
10%	93.95	93.53	93.65	93.83	93.29	77.75	98.11	98.05	98.02	98.16	98.04	92.25
20%	92.77	92.25	92.63	92.23	92.67	74.75	97.66	97.63	97.67	97.49	97.34	90.56
30%	90.85	91.25	91.34	91.39	90.99	70.94	96.74	96.90	96.67	96.63	96.90	87.97
40%	89.60	89.04	89.06	89.50	89.32	66.10	95.83	95.62	95.52	95.64	95.66	84.33

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

Table 2: Coverage (%) − Non−shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5

	None adjustment						Many−to−one adjustment					
Missing rate	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**Joint**	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**Joint**
0%	94.76	95.30	95.31	95.23	95.12	83.01	98.38	98.57	98.49	98.44	98.57	94.07
Imputation using linear regression												
10%	94.64	94.73	94.51	94.75	94.62	81.26	98.48	98.62	98.51	98.41	98.41	93.86
20%	95.19	94.95	95.10	94.82	95.36	82.72	98.66	98.51	98.67	98.51	98.80	94.59
30%	95.18	95.57	95.47	95.89	95.48	83.32	98.80	99.06	98.83	98.97	98.86	95.53
40%	95.79	95.87	95.69	95.88	95.73	84.70	98.86	98.79	98.74	99.19	98.92	95.39
Imputation using predictive mean matching method												
10%	93.58	93.72	93.44	93.72	93.65	78.11	98.12	98.08	98.08	97.85	98.06	92.17
20%	92.94	92.87	92.71	92.70	93.28	76.06	97.72	97.55	97.74	97.63	97.86	91.30
30%	91.75	91.42	91.49	92.06	91.66	71.77	96.91	97.27	96.89	97.26	97.29	89.04
40%	89.40	89.71	89.44	89.43	89.47	66.75	95.65	95.83	95.74	95.85	96.00	84.82
Imputation using propensity−score method												
10%	93.71	93.65	93.34	93.54	93.54	78.00	98.17	98.00	98.08	97.86	97.93	92.10
20%	92.63	92.53	92.78	92.53	93.16	75.40	97.56	97.39	97.51	97.36	97.82	90.67
30%	90.72	91.24	91.28	91.31	91.03	70.37	96.49	96.99	96.70	96.88	96.93	87.83
40%	89.14	89.00	88.68	88.88	88.92	64.93	95.46	95.46	95.47	95.44	95.31	83.48

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

Table 3: Average interval length (expansion %) − Non−shrinkage

Scenario I: all means are 0 for T0,T1,...,T5

	None adjustment					Many−to−one adjustment				
Missing rate	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**
0%	4.028	4.075	4.108	4.147	4.186	5.034	5.093	5.135	5.183	5.233
Imputation using linear regression										
10%	4.176 (3.67)	4.213 (3.39)	4.256 (3.60)	4.288 (3.42)	4.338 (3.63)	5.320 (5.68)	5.367 (5.39)	5.422 (5.59)	5.464 (5.40)	5.527 (5.61)
20%	4.539 (12.68)	4.573 (12.24)	4.617 (12.37)	4.662 (12.44)	4.711 (12.54)	5.781 (14.85)	5.825 (14.37)	5.881 (14.51)	5.939 (14.57)	6.001 (14.67)
30%	5.045 (25.23)	5.121 (25.68)	5.147 (25.29)	5.204 (25.50)	5.253 (25.48)	6.420 (27.52)	6.517 (27.96)	6.551 (27.56)	6.623 (27.77)	6.685 (27.75)
40%	5.825 (44.60)	5.855 (43.69)	5.909 (43.83)	5.983 (44.29)	6.056 (44.67)	7.405 (47.10)	7.443 (46.15)	7.512 (46.29)	7.607 (46.76)	7.700 (47.14)
Imputation using predictive mean matching method										
10%	3.997 (−0.78)	4.034 (−1.01)	4.074 (−0.83)	4.107 (−0.95)	4.154 (−0.77)	5.093 (1.17)	5.140 (0.92)	5.192 (1.10)	5.234 (0.98)	5.294 (1.16)
20%	4.110 (2.02)	4.143 (1.68)	4.183 (1.82)	4.224 (1.87)	4.266 (1.91)	5.236 (4.01)	5.279 (3.66)	5.330 (3.80)	5.382 (3.84)	5.436 (3.88)
30%	4.235 (5.12)	4.286 (5.17)	4.321 (5.18)	4.359 (5.11)	4.403 (5.18)	5.394 (7.14)	5.458 (7.18)	5.504 (7.18)	5.552 (7.10)	5.608 (7.17)
40%	4.376 (8.64)	4.413 (8.30)	4.450 (8.31)	4.500 (8.52)	4.549 (8.67)	5.572 (10.68)	5.619 (10.33)	5.666 (10.33)	5.730 (10.54)	5.793 (10.69)
Imputation using propensity−score method										
10%	4.007 (−0.52)	4.042 (−0.81)	4.084 (−0.60)	4.117 (−0.71)	4.164 (−0.53)	5.106 (1.43)	5.150 (1.12)	5.203 (1.32)	5.246 (1.21)	5.306 (1.39)
20%	4.137 (2.68)	4.169 (2.32)	4.211 (2.50)	4.250 (2.50)	4.294 (2.56)	5.269 (4.66)	5.310 (4.27)	5.364 (4.45)	5.414 (4.45)	5.469 (4.51)
30%	4.264 (5.84)	4.316 (5.93)	4.354 (5.97)	4.393 (5.93)	4.434 (5.91)	5.427 (7.80)	5.494 (7.87)	5.542 (7.92)	5.591 (7.87)	5.644 (7.85)
40%	4.365 (8.37)	4.401 (8.01)	4.447 (8.24)	4.496 (8.43)	4.548 (8.64)	5.554 (10.32)	5.599 (9.94)	5.658 (10.17)	5.721 (10.37)	5.786 (10.57)

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

Table 3: Average interval length (expansion %) − Non−shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5

	None adjustment					Many−to−one adjustment				
Missing rate	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**	**T1vs.T0**	**T2vs.T0**	**T3vs.T0**	**T4vs.T0**	**T5vs.T0**
0%	4.036	4.077	4.119	4.152	4.191	5.042	5.094	5.147	5.189	5.238
Imputation using linear regression										
10%	4.160 (3.07)	4.203 (3.09)	4.245 (3.08)	4.292 (3.38)	4.327 (3.25)	5.301 (5.13)	5.356 (5.14)	5.410 (5.12)	5.470 (5.42)	5.514 (5.28)
20%	4.545 (12.61)	4.600 (12.83)	4.636 (12.57)	4.692 (13.00)	4.732 (12.91)	5.787 (14.77)	5.857 (14.97)	5.903 (14.70)	5.974 (15.13)	6.025 (15.04)
30%	5.035 (24.75)	5.084 (24.69)	5.129 (24.54)	5.191 (25.03)	5.227 (24.72)	6.411 (27.14)	6.473 (27.06)	6.531 (26.90)	6.611 (27.40)	6.656 (27.08)
40%	5.821 (44.21)	5.885 (44.35)	5.921 (43.76)	5.978 (43.98)	6.040 (44.12)	7.397 (46.70)	7.480 (46.83)	7.525 (46.22)	7.598 (46.43)	7.677 (46.58)
Imputation using predictive mean matching method										
10%	3.986 (−1.24)	4.028 (−1.20)	4.067 (−1.25)	4.113 (−0.94)	4.145 (−1.10)	5.080 (0.74)	5.134 (0.78)	5.183 (0.71)	5.242 (1.02)	5.283 (0.86)
20%	4.119 (2.06)	4.165 (2.15)	4.198 (1.94)	4.249 (2.33)	4.284 (2.22)	5.246 (4.05)	5.305 (4.14)	5.348 (3.90)	5.412 (4.31)	5.457 (4.19)
30%	4.234 (4.90)	4.277 (4.91)	4.311 (4.67)	4.367 (5.17)	4.396 (4.89)	5.394 (6.97)	5.449 (6.97)	5.492 (6.72)	5.564 (7.23)	5.601 (6.93)
40%	4.354 (7.88)	4.404 (8.02)	4.446 (7.95)	4.485 (8.02)	4.527 (8.03)	5.545 (9.97)	5.609 (10.10)	5.662 (10.02)	5.712 (10.08)	5.766 (10.10)
Imputation using propensity−score method										
10%	3.997 (−0.97)	4.039 (−0.94)	4.077 (−1.01)	4.124 (−0.68)	4.156 (−0.82)	5.093 (1.01)	5.146 (1.02)	5.195 (0.94)	5.255 (1.28)	5.297 (1.13)
20%	4.145 (2.71)	4.189 (2.73)	4.226 (2.60)	4.278 (3.03)	4.310 (2.85)	5.278 (4.67)	5.333 (4.69)	5.380 (4.53)	5.447 (4.97)	5.488 (4.79)
30%	4.263 (5.61)	4.310 (5.70)	4.346 (5.51)	4.401 (6.00)	4.428 (5.66)	5.426 (7.62)	5.486 (7.70)	5.533 (7.50)	5.604 (8.00)	5.638 (7.64)
40%	4.357 (7.96)	4.404 (8.02)	4.465 (8.41)	4.492 (8.18)	4.541 (8.35)	5.544 (9.96)	5.604 (10.01)	5.682 (10.40)	5.716 (10.16)	5.779 (10.34)

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv